

Verra VM0051 Tier 2 - AWD Monitoring Report

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1. Field Information

Field ID:	new1
Field Name:	Rangpur Farm
District:	Rangpur
Area:	6450.6293 ha

2. Monitoring Period

Season:	
Window Start:	2026-01-01
Window End:	2026-05-31
Satellite:	Sentinel-1 SAR GRD, IW mode, DESCENDING pass
Bands Used:	VV + VH (C-band, 10 m native resolution)
Spatial Reducer:	Median over field polygon (robust to speckle)

3. Satellite Data Summary

Total Observations:	12
VV Mean:	-12.6741 dB
VV Std Dev:	0.5171 dB
Smoothing Applied:	Savitzky-Golay (window=5, polyorder=2)
Flood Criterion:	Z-score < -0.8 (field-adaptive baseline)
Drydown Criterion:	dVV > 1.2 sigma after flooded state

4. AWD Events and Phenology

AWD Cycles Detected:	2
Drydown Dates:	2026-03-10, 2026-04-27
Sowing Date:	2026-02-26
Harvest Date:	2026-05-21
Season Length:	84 days (detected from VH phenology signal)

5. Carbon Estimation (Verra VM0051 v1.1, QA3 - Default Emission Factors)

Quantification Approach:	QA3 - Default Emission Factors (§8.2.3)
Emission Factor EF_c:	0.85 kg CH4/ha/day (IPCC 2019 Table 5.11, South Asia Tier 1 regional default)
GWP CH4 (AR5 100yr):	28
GWP N2O (AR5 100yr):	265
Baseline SF_w:	1.00 (continuous flooding, Eq. 6)
Project SF_w:	0.55 (Eq. 6, >=2 drydowns)

Pre-season SC _p :	1.0 (Eq. 6, Table 5.13)
Organic amendment SC _o (baseline):	2.8781 (Eq. 7, Table 5.14)
Organic amendment SC _o (project):	2.8781 (Eq. 7, Table 5.14)
Baseline CH ₄ Emissions:	1325590.9621 kg CH ₄ (Eq. 6/8)
Project CH ₄ Emissions:	729075.0292 kg CH ₄ (Eq. 6/8)
Gross CH ₄ Avoided:	596515.9330 kg CH ₄
Gross tCO ₂ e (before UNC):	16702.446123 tCO ₂ e
Uncertainty Deduction:	15.0% (QA3 flat rate, §8.6.3, ≤60,000 tCO ₂ e/yr)
UNC Deduction Amount:	2505.366918 tCO ₂ e
CH ₄ After Uncertainty:	14197.079204 tCO ₂ e
N Fertilizer Input (Q _N):	100.0 kg N/ha
N ₂ O Correction (Eq. 25):	536.756864 tCO ₂ e (PE_Red-Irri, CF _{N2O} =0.00314)
N ₂ O Penalty (% of gross, informational):	2% (mandatory Eq. 29 term, not a §8.4 leakage screen)
NET ISSUANCE (Eq. 29):	13660.322340 tCO ₂ e

CALCULATED ESTIMATE: 13660.3223 tCO₂e (not registry-issued credits) - ready for registry submission.

6. Methodology

Terra-Audit implements the Verra VM0051 v1.1 Quantification Approach 3 (QA3 - Default Emission Factors) for Alternate Wetting and Drying (AWD) rice irrigation monitoring using Sentinel-1 SAR satellite data. VV backscatter is used as the primary flood-state indicator via z-score anomaly detection (field-adaptive baseline, threshold $z < -0.8$). Sharp positive VV transitions (> 1.2 sigma) following flooded periods are classified as drydown events. The VM0051 water scaling factor (SF_w) is assigned per Eq. 6/Table 5.12: 1.00 (0 events), 0.71 (1 event), 0.55 (≥ 2 events), alongside a pre-season water-regime factor (SC_p, Table 5.13) and an organic-amendment factor (SC_o, Eq. 7/Table 5.14). Gross CH₄ reductions are reduced by the QA3 flat 15% uncertainty deduction (§8.6.3, valid only at or below the 60,000 tCO₂e/yr project-size gate) and by the N₂O irrigation correction (PE_Red-Irri, §8.3.2 Eq. 25, CF_{N2O}=0.00314 kg N₂O/kg N) -- a mandatory project emission subtracted in full, not a §8.4 leakage source. Net reductions follow Eq. 29 (simplified: CH₄ soil term only; §8.4 leakage sources such as organic-amendment import and yield decline are not computed by this engine).

7. Assumptions

1. EF_c = 0.85 kg CH₄/ha/day (IPCC 2019 Table 5.11, South Asia Tier 1 regional default; may not reflect site-specific soil conditions)
2. DESCENDING orbit pass only used to avoid time-series artefacts from mixed-orbit acquisition geometry
3. Field area computed via Shoelace formula with spherical latitude correction (no external GIS dependency)
4. Single cropping season assumed per analysis window
5. Pre-season water regime (SC_p) and organic-amendment (SC_o) factors are user-entered (default: 5 t/ha straw, incorporated shortly before cultivation, per §8.2.3 footnote 16) -- verify against actual farm practice
6. Soil type / rice cultivar scaling (SFs/SFr) not modeled -- no site-specific soil characterization data incorporated

8. Limitations

1. SAR backscatter is sensitive to surface roughness and wind -- drydown events may be missed or false-positive during high-wind conditions
2. Flooding under dense crop canopy may be underestimated by VV backscatter
3. Phenology detection requires ≥ 5 observations and a discernible VH signal minimum; fails silently on short or noisy windows
4. Results have not been validated against in-situ soil moisture sensors or farmer irrigation logbooks
5. This report is generated by an AI-assisted analytical platform and requires expert agronomist review before formal registry submission